

ECON 3350/7350 Applied Econometrics for Macroeconomics and Finance

Lecture 7: Volatility Models (1)

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Volatility Models

Recommended readings

Author	Title	Chapter	Call No
Enders	Applied Econometric Time Series, 4e	3	HB139 .E55 2015
Verbeek	A Guide to Modern Econometrics	8.11	HB139 .V465 2012

Features of Financial Data

Linear models cannot explain a number of important features common to financial data:

- **leprokurtosis** or “fat tails”: high negative returns that have higher-than-expected probability in historic time-series;
- **volatility clustering** or volatility “pooling”: big shocks tend to follow big shocks (either direction) and small shocks tend to follow small shocks;
- **leverage effects** or volatility “asymmetry”: for example, volatility of a stock price tends to increase when its price drops (Black, 1976).

Typical Example of Returns

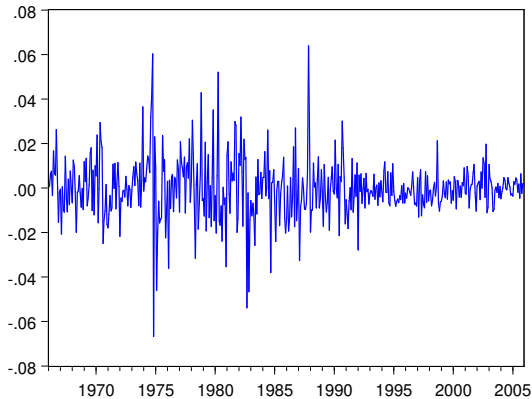


Figure 1: Returns to dividend yield S&P 500 index: 1996-2005 Monthly.

Volatility

In the classical regression model:

$$y_i = \beta_1 + \beta_2 x_{2,i} + \cdots + \beta_k x_{k,i} + \varepsilon_i,$$

we assumed $\text{Var}(\varepsilon_i) = \sigma^2$.

In financial time-series, we observe large variations in **volatility**.

Volatility is related to the **variance** of a process, **not** the mean.

Volatility is a form of **heteroskedasticity**; in this case, **conditional variance** is modeled as a function of time.

Non-linear Models

Consider the **non-linear** stochastic process (Campbell, Lo and MacKinlay, 1997):

$$y_t = \underbrace{g(\varepsilon_{t-1}, \varepsilon_{t-2}, \dots)}_{\text{conditional mean}} + \nu_t \underbrace{h^{\frac{1}{2}}(\varepsilon_{t-1}, \varepsilon_{t-2}, \dots)}_{\text{volatility}}.$$

Models with **non-linear** $g(\cdot)$ are **non-linear in mean**.

Models with **non-constant** $h(\cdot)$ are **non-linear in variance**.

We will focus on **linear** $g(\cdot)$, but **non-constant** $h(\cdot)$.

Autoregressive Conditional Heteroskedasticity (ARCH)

Simplest example was proposed by Engle (1982):

$$h_t \equiv E(\varepsilon_t^2 | \varepsilon_{t-1}, \varepsilon_{t-2}, \dots) = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2.$$

This is called the **Auto-Regressive Conditional Heteroskedasticity** of order 1 model, or ARCH(1).

To ensure that $h_t \geq 0$ regardless of ε_{t-1}^2 , we impose $\alpha_0 \geq 0$ and $\alpha_1 \geq 0$.

ARCH(1) captures the idea that when a **large shock** happens in $t - 1$, there is a greater probability of larger shocks in t .

- For larger ε_{t-1} , the next shock ε_t is still **mean-zero** but has **larger variance** (proportional to ε_{t-1}^2).

ARCH(1)

The ARCH(1) specification indicates ε_t^2 and ε_{t-1}^2 are correlated.

However, the **unconditional** variance of ε_t is **homoskedastic**:

$$E(\varepsilon_t^2) = \alpha_0 + \alpha_1 E(\varepsilon_{t-1}^2).$$

For $0 \leq \alpha_1 < 1$, the stationary solution is

$$E(\varepsilon_t^2) = \frac{\alpha_0}{1 - \alpha_1}.$$

ARCH(q)

The ARCH(1) can be easily extended to an ARCH(q) process:

$$\begin{aligned}h_t &\equiv E(\varepsilon_t^2 \mid \varepsilon_{t-1}, \varepsilon_{t-2}, \dots) = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \alpha_2 \varepsilon_{t-2}^2 + \dots + \alpha_q \varepsilon_{t-q}^2, \\&= \alpha_0 + \alpha(L) \varepsilon_{t-1}^2,\end{aligned}$$

where $\alpha(L) = \alpha_1 + \alpha_2 L + \dots + \alpha_q L^{q-1}$ is a lag polynomial of order $q - 1$.

All coefficients are non-negative, i.e. $\alpha_j \geq 0$ for $j = 0, \dots, q$.

To ensure stationarity, impose $\alpha(1) < 1$.

In an ARCH(q), only the past q shocks affect the current volatility.

Combining Linear Models with ARCH Errors

We can specify a regression with ARCH(1) errors as:

$$y_t = \beta_1 + \beta_2 x_{2,t} + \cdots + \beta_k x_{k,t} + \varepsilon_t,$$
$$\varepsilon_t = \nu_t \sqrt{\alpha_0 + \alpha_1 \varepsilon_{t-1}^2},$$

where $\nu_t \sim \mathcal{N}(0, 1)$ and ν_t is independent of ε_{t-s} for all t, s .

We can specify an AR(1) model with ARCH(2) errors as:

$$y_t = a_0 + a_1 y_{t-1} + \varepsilon_t,$$
$$\varepsilon_t = \nu_t \sqrt{\alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \alpha_2 \varepsilon_{t-2}^2},$$

where $\nu_t \sim \mathcal{N}(0, 1)$ and ν_t is independent of ε_{t-s} for all t, s .

Examples of Models with ARCH Errors

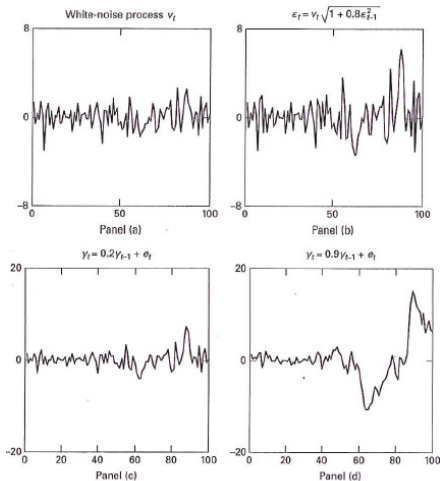


FIGURE 3.7 Simulated ARCH Processes

Source: Enders (2015)

Consequence of ARCH Errors

The presences of ARCH errors **does not** invalidate OLS, but there exist **more efficient** non-linear estimators to estimate **conditional-mean** coefficients.

If the objective is to **forecast volatility**, then it is relevant to **test** for the presence of ARCH errors and **estimate** an ARCH model.

Breusch-Pagan type test for heteroskedasticity: $H_0 : \alpha_1 = \dots = \alpha_q = 0$ versus $H_1 : \text{ARCH}(q)$.

1. **Estimate** the conditional mean using OLS and save residuals in $\{\hat{\varepsilon}_t\}$.
2. **Regress** $\hat{\varepsilon}_t^2 = \alpha_0 + \alpha_1 \hat{\varepsilon}_{t-1}^2 + \dots + \alpha_q \hat{\varepsilon}_{t-q}^2$ and compute R_e^2 .
3. **Test** using $\text{LM}_{\text{ARCH}} \equiv TR_e^2 \sim \chi_q^2$.

Generalized ARCH (GARCH)

The **Generalized** ARCH model (proposed by Bollerslev, 1986) is:

$$\begin{aligned}\varepsilon_t^2 &= \nu_t^2 h_t, \\ h_t &= \alpha_0 + \sum_{j=1}^q \alpha_j \varepsilon_{t-j}^2 + \sum_{j=1}^p \beta_j h_{t-j}, \\ &= \alpha_0 + \alpha(L) \varepsilon_{t-1}^2 + \beta(L) h_{t-1},\end{aligned}$$

known as GARCH(p, q).

Again, we require $\alpha_j \geq 0$ for $j = 0, \dots, q$ and $\beta_j \geq 0$ for $j = 1, \dots, p$ to ensure $h_t \geq 0$.

In practice, the most commonly used is the GARCH(1, 1):

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1}.$$

GARCH(1, 1)

Define the **surprise in the squared innovation** $\eta_t = \varepsilon_t^2 - h_t$, and re-write the GARCH(1, 1) as an ARMA(1, 1):

$$\varepsilon_t^2 = \nu_t^2 h_t,$$

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1},$$

and adding / subtracting $\beta_1 \varepsilon_{t-1}^2, \varepsilon_t^2$ yields

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1} + (\varepsilon_t^2 - \varepsilon_t^2) + (\beta_1 \varepsilon_{t-1}^2 - \beta_1 \varepsilon_{t-1}^2),$$

$$\varepsilon_t^2 = \alpha_0 + (\alpha_1 + \beta_1) \varepsilon_{t-1}^2 - \beta_1 (\varepsilon_{t-1}^2 - h_{t-1}) + (\varepsilon_t^2 - h_t),$$

$$\varepsilon_t^2 = \alpha_0 + (\alpha_1 + \beta_1) \varepsilon_{t-1}^2 + \eta_t - \beta_1 \eta_{t-1}.$$

The “error” η_t is uncorrelated over time, but heteroskedastic.

GARCH(1, 1)

The autoregressive coefficient $\alpha_1 + \beta_1$, so **stationarity** requires $\alpha_1 + \beta_1 < 1$; values of $\alpha_1 + \beta_1$ close to 1 indicate **highly persistent** volatility.

Under stationarity, the **unconditional variance** of ε_t is:

$$E(\varepsilon_t^2) = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}.$$

Similar to AR and MA processes, The GARCH(1, 1) can be written as an **infinite-order** ARCH model with **geometrically declining coefficients**:

$$\begin{aligned} h_t &= \alpha_0(1 + \beta_1 + \beta_1^2 + \cdots) + \alpha_1(\varepsilon_{t-1}^2 + \beta_1\varepsilon_{t-2}^2 + \beta_1^2\varepsilon_{t-3}^2 + \cdots), \\ &= \frac{\alpha_0}{1 - \beta_1} + \alpha_1 \sum_{j=1}^{\infty} \beta_1^{j-1} \varepsilon_{t-j}^2. \end{aligned}$$

The GARCH(1, 1) is a **parsimonious** alternative to a high-order ARCH.

Example

Linear regression to forecast S&P 500 excess returns (dependent variable EXRET).

Explanatory Variables	
CS_1	credit spread (yield on Moody's Aaa minus BBa debt), lagged one month
DY_1	dividend yield S&P 500 index, lagged one month (in % per month)
I12_1	12-month interest rate, lagged one month
I12_2	12-month interest rate, lagged two months
I3_1	1-month interest rate, lagged one month
I3_2	1-month interest rate, lagged two months
INF_2	inflation, lagged two months
IP_2	change industrial production, lagged two months
MB_2	change in monetary base, lagged two months
PE_1	price earnings ratio (S&P500), lagged one month
WINTER	dummy, 1 in November to April, 0 otherwise

Estimation Results

Dependent Variable: EXRET

Method: Least Squares

Date: 04/14/16 Time: 16:10

Sample: 1966M01 2005M12

Included observations: 480

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	3.300976	2.325907	1.419221	0.1565
PE.1	-11.39836	6.450444	-1.767067	0.0779
DY.1	5.989411	6.011935	0.996253	0.3196
INF.2	-13.98802	6.866371	-2.037178	0.0422
IP.2	-2.111113	5.694212	-0.370747	0.7110
I3.1	19.18752	12.11589	1.583666	0.1139
I3.2	-19.45177	11.96867	-1.625224	0.1048
I12.1	-41.34318	12.08628	-3.420670	0.0007
I12.2	35.89172	12.55039	2.859809	0.0044
MB.2	-12.85298	6.159389	-2.086729	0.0375
CS.1	18.30060	9.921162	1.844602	0.0657
WINTER	0.805715	0.391515	2.057941	0.0401
R-squared	0.117856	Mean dependent var		0.432739
Adjusted R-squared	0.097122	S.D. dependent var		4.382373
S.E. of regression	4.164128	Akaike info criterion		5.715573
Sum squared resid	8115.100	Schwarz criterion		5.819917
Log likelihood	-1359.737	Hannan-Quinn criter.		5.756588
F-statistic	5.684130	Durbin-Watson stat		2.076803
Prob(F-statistic)	0.000000			

Residuals ACF/PACF

Sample: 1966M01 2006M12

Included observations: 480

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 -0.039	-0.039	0.7182	0.397
		2 -0.016	-0.018	0.8473	0.655
		3 0.039	0.037	1.5679	0.667
		4 -0.041	-0.038	2.3705	0.668
		5 0.096	0.094	6.8308	0.234
		6 -0.000	0.004	6.8308	0.337
		7 0.042	0.049	7.6779	0.362
		8 -0.000	-0.006	7.6780	0.466
		9 0.024	0.033	7.9529	0.539
		10 0.058	0.048	9.6071	0.476
		11 0.000	0.009	9.6071	0.566
		12 0.037	0.030	10.293	0.590
		13 0.006	0.007	10.309	0.668
		14 -0.045	-0.048	11.328	0.660
		15 -0.005	-0.020	11.339	0.728
		16 0.012	0.008	11.413	0.783
		17 0.080	0.075	14.645	0.621
		18 0.014	0.015	14.736	0.680

Squared Residuals ACF/PACF

Sample: 1966M01 2006M12

Included observations: 480

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob	
		1	0.144	0.144	9.9571	0.002
		2	0.101	0.083	14.941	0.001
		3	0.122	0.100	22.179	0.000
		4	0.018	-0.019	22.345	0.000
		5	0.026	0.007	22.662	0.000
		6	0.012	-0.004	22.738	0.001
		7	-0.055	-0.060	24.210	0.001
		8	-0.026	-0.016	24.543	0.002
		9	0.080	0.097	27.651	0.001
		10	0.083	0.081	31.002	0.001
		11	0.018	-0.012	31.156	0.001
		12	0.007	-0.026	31.183	0.002
		13	0.079	0.067	34.268	0.001
		14	0.041	0.019	35.083	0.001
		15	0.095	0.075	39.565	0.001
		16	0.025	-0.006	39.883	0.001
		17	-0.012	-0.019	39.949	0.001
		18	0.004	-0.017	39.958	0.002

GARCH(1, 1) Estimation Results

Dependent Variable: EXRET

Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	2.117416	2.480921	0.853480	0.3934
PE.1	-7.263156	7.102047	-1.022685	0.3065
DY.1	9.306910	5.394256	1.725337	0.0845
INF.2	-11.79318	6.239101	-1.890206	0.0587
IP.2	-4.637199	5.526029	-0.839156	0.4014
I3.1	26.71094	9.239231	2.891035	0.0038
I3.2	-28.12567	9.851971	-2.854826	0.0043
I12.1	-49.69434	10.65276	-4.664927	0.0000
I12.2	45.13972	11.28990	3.998238	0.0001
MB.2	-15.51522	4.579737	-3.387797	0.0007
CS.1	15.54553	10.58376	1.468809	0.1419
WINTER	0.793297	0.375668	2.111698	0.0347
Variance Equation				
C	1.078377	0.576406	1.870863	0.0614
RESID(-1) ²	0.108658	0.041255	2.633798	0.0084
GARCH(-1)	0.832238	0.057704	14.42264	0.0000
R-squared	0.114786	Mean dependent var		0.432739
Adjusted R-squared	0.093979	S.D. dependent var		4.382373
S.E. of regression	4.171367	Akaike info criterion		5.671065
Sum squared resid	8143.342	Schwarz criterion		5.801495
Log likelihood	-1346.055	Hannan-Quinn criter.		5.722334
Durbin-Watson stat	2.064821			

GARCH(1, 1) Residuals ACF/PACF

Date: 09/05/16 Time: 23:52

Sample: 1966M01 2005M12

Included observations: 480

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob*
		1 -0.045	-0.045	0.9832	0.321
		2 0.015	0.013	1.0869	0.581
		3 0.048	0.049	2.1947	0.533
		4 -0.037	-0.033	2.8459	0.584
		5 0.096	0.092	7.2982	0.199
		6 -0.019	-0.013	7.4798	0.279
		7 0.043	0.043	8.3919	0.299
		8 0.008	0.002	8.4262	0.393
		9 0.054	0.062	9.8343	0.364
		10 0.051	0.043	11.138	0.347
		11 0.005	0.014	11.152	0.431
		12 0.031	0.018	11.617	0.477
		13 -0.006	-0.004	11.632	0.558
		14 -0.019	-0.030	11.803	0.622
		15 -0.007	-0.019	11.829	0.692
		16 -0.008	-0.012	11.859	0.754
		17 0.087	0.081	15.643	0.549
		18 0.012	0.016	15.712	0.613

GARCH(1, 1) Squared Residuals ACF/PACF

Date: 09/05/16 Time: 23:53

Sample: 1966M01 2005M12

Included observations: 480

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob*
		1 -0.037	-0.037	0.6589	0.417
		2 0.017	0.015	0.7929	0.673
		3 0.061	0.062	2.5731	0.462
		4 -0.024	-0.020	2.8463	0.584
		5 -0.021	-0.025	3.0590	0.691
		6 -0.011	-0.016	3.1227	0.793
		7 -0.091	-0.089	7.2083	0.408
		8 -0.040	-0.045	8.0105	0.432
		9 0.051	0.052	9.2635	0.413
		10 0.043	0.060	10.168	0.426
		11 -0.022	-0.019	10.399	0.495
		12 -0.022	-0.038	10.629	0.561
		13 0.048	0.038	11.772	0.546
		14 0.007	0.009	11.796	0.623
		15 0.082	0.081	15.102	0.444
		16 0.022	0.032	15.351	0.499
		17 -0.031	-0.021	15.825	0.536
		18 -0.003	-0.019	15.829	0.604

Forecasting Volatility from GARCH(p, q)

Consider a model with GARCH errors and volatility given by

$$h_{t+1} = \alpha_0 + \alpha_1 \varepsilon_t^2 + \beta_1 h_t.$$

Assuming the parameters are known, the **one-step ahead** forecast is given by:

$$\hat{h}_{t+1} = E(h_{t+1} | \varepsilon_t, \varepsilon_{t-1}, \dots) = \alpha_0 + \alpha_1 \varepsilon_t^2 + \beta_1 h_t$$

Assuming the parameters are known, the **j -step ahead** forecast for $j > 1$ is given by:

$$\begin{aligned} E(h_{t+j} | \varepsilon_t, \varepsilon_{t-1}, \dots) &= \alpha_0 + \alpha_1 E(\varepsilon_{t+j-1}^2 | \varepsilon_t, \varepsilon_{t-1}, \dots) \\ &\quad + \beta_1 E(h_{t+j-1} | \varepsilon_t, \varepsilon_{t-1}, \dots), \\ &= \alpha_0 + (\alpha_1 + \beta_1) E(h_{t+j-1} | \varepsilon_t, \varepsilon_{t-1}, \dots), \end{aligned}$$

since $E(\varepsilon_{t+j-1}^2 | \varepsilon_t, \varepsilon_{t-1}, \dots) = E(h_{t+j-1} | \varepsilon_t, \varepsilon_{t-1}, \dots)$.

Forecasting Volatility from GARCH(p, q)

We can write this more compactly as

$$\begin{aligned} E_t(h_{t+j}) &= (\alpha_1 + \beta_1)E_t(h_{t+j-1}), \\ &= \alpha_0 (1 + (\alpha_1 + \beta_1) + (\alpha_1 + \beta_1)^2 + \cdots + (\alpha_1 + \beta_1)^{j-1}) \\ &\quad + (\alpha_1 + \beta_1)^j h_t. \end{aligned}$$

If $\alpha_1 + \beta_1 < 1$, the conditional forecast of h_{t+j} will converge as $j \rightarrow \infty$ to the **long-run** value:

$$\lim_{j \rightarrow \infty} E_t(h_{t+j}) = E(h_t) = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}.$$